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URAG:Unified Retrieval-Augmented Generation

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Abstract

To address the issues of insufficient retrieval capabilities and "hallucinations" in responses generated by large models, this paper proposes a knowledge question-answering framework based on Unified Retrieval-Augmented Generation (URAG). The framework integrates three retrieval mechanisms—keyword retrieval, vector retrieval, and graph retrieval—enabling efficient and high-quality utilization of massive, multi-source, and heterogeneous data. It effectively overcomes the limitations of a single retrieval pathway. Experimental results demonstrate that the URAG framework excels across various task scenarios, enhancing the accuracy and comprehensiveness of knowledge retrieval. Its advantage is particularly evident when dealing with multi-dimensional and multi-layered information. In conclusion, the Unified Retrieval-Augmented Generation technique offers a new method for improving the performance of intelligent question-answering systems, with broad application prospects and research value. This study provides valuable insights into understanding RAG (Retrieval-Augmented Generation) technology and its application in large language models.

CCS Concepts

• **Computing methodologies → Natural language generation**.

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Keywords

Unified Retrieval-Augmented Generation (URAG),Multi-source heterogeneous data,Accuracy and comprehensiveness in knowledge retrieval

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1 Introduction

Question-answering technology has long been a key research focus in the fields of natural language processing (NLP) and information retrieval. Its goal is 'to enable computers to answer questions posed by users in natural language with accurate and concise natural language responses.' The rise of this research is driven by the demand for quick and accurate access to information. Large models have enhanced the relevance and fluency of generated answers through instruction fine-tuning and reinforcement feedback[6, 53, 54]. For instance, InstructGPT significantly improved the alignment of model outputs with human expectations by using instruction fine-tuning strategies, leading to more coherent and contextually appropriate responses[34]. However, the generally low retrieval capabilities of large models have reduced the reliability and traceability of their answers[31, 33]. For example, a recent study showed that state-of-the-art large language models (LLMs) had a hallucination rate of 69% to 88% when dealing with specific legal queries[9].

Retrieval-Augmented Generation (RAG) technology, first proposed by Lewis et al., addresses the shortcomings of large language models in query processing, such as the 'hallucination' problem that arises when handling queries in specific or highly specialized domains[24, 25, 45, 50]. Specifically, RAG first invokes a retriever to search and extract relevant documents from an external database and then uses these documents as context to enhance the

generation process[18]. In practice, RAG technology is both feasible and efficient, as it can be applied to various generation tasks by simply adjusting the retrieval component, requiring little to no additional training[40]. It has demonstrated great potential, particularly in knowledge-intensive tasks such as open-domain question answering (OpenQA)[2, 16, 38, 44].

In the current era of information, data resources are vast, multi-sourced, and heterogeneous. Existing single-path retrieval methods, such as vector retrieval, keyword retrieval, or graph retrieval, struggle to efficiently and effectively utilize large-scale, multi-source, heterogeneous data. When facing complex semantic queries or questions with diverse expressions, relying solely on keyword retrieval may miss relevant information, especially when the query keywords are inaccurate or inconsistent with the target text[14]. Vector retrieval performs well at the semantic level but can sometimes over-generalize, leading to a decrease in the relevance of retrieval results, especially when semantic boundaries are ambiguous or low-frequency terms are overlooked[22, 42]. Graph retrieval effectively handles structured data and complex relationships but is limited in its utility when dealing with unstructured information[37, 39].

To address the aforementioned issues, this paper proposes a knowledge question-answering framework based on Unified Retrieval-Augmented Generation (URAG). This framework is composed of three main components: knowledge storage, knowledge retrieval, and fusing retrieval results. In the knowledge storage component, all text content is extracted and converted into indexed formats for storage. In the knowledge retrieval component, the system retrieves the most relevant textual knowledge from the knowledge base according to the user's input query. In the stage of fusing retrieval results, the outputs from the three retrieval paths are merged and processed. The results are optimized through operations such as re-ranking and filtering, and then the user's question and the fusing retrieval results are input into the language model to generate the final response.

The contributions of this work are summarized as follows:

1. A technology framework based on Unified Retrieval-Augmented Generation (URAG) is proposed. The knowledge retrieval phase introduces a three-path retrieval mechanism: keyword retrieval, vector retrieval, and graph retrieval. The synergistic effect of these three retrieval methods not only precisely matches keywords but also deeply understands data at the semantic and relational levels. This compensates for the shortcomings of single-path methods and provides technical support for future intelligent generation systems.

2. Multiple experiments conducted on datasets across various task scenarios have validated the accuracy and comprehensiveness of this technical framework in knowledge retrieval, particularly in the process of handling multi-dimensional and multi-level information.

2 Related Work

2.1 Development of Question-Answering Systems

Question-answering (QA) systems have long been a central focus in the fields of natural language processing (NLP) and information retrieval (IR) [23]. Their goal is to generate accurate and relevant answers to user queries posed in natural language [17]. Early QA systems primarily relied on structured databases and predefined rules to provide answers [47]. These systems were largely based on keyword matching and template filling, which limited their ability to handle complex, open-domain questions [49].

In recent years, with the advent of deep learning and large language models (LLMs) [48], QA systems have seen significant performance improvements. Pre-trained models like BERT and GPT-3 are trained on massive amounts of text data and fine-tuned for specific tasks, leading to better handling of complex natural language queries [3, 10, 35]. Models such as InstructGPT [36], which use instruction fine-tuning, have further improved the relevance and coherence of the generated responses in open-domain QA tasks. However, these models still face issues with hallucination, especially in knowledge-intensive domains like law and medicine [20, 28]. Studies have shown that hallucination rates in current large language models (LLMs) can range from 69% to 88% when handling specific domain queries [8, 9, 27, 51].

2.2 Retrieval-Augmented Generation (RAG) Technology

To address the hallucination problem in large models, Retrieval-Augmented Generation (RAG) was proposed by Lewis et al. [1, 24]. The core idea of RAG is to retrieve relevant documents from external knowledge bases before generating the final answer, thus improving the accuracy of the generated responses. By combining retrieval and generation, RAG allows models to dynamically access external knowledge, reducing the likelihood of hallucinations.

RAG has shown great potential in knowledge-intensive tasks, especially in open-domain QA. Chen et al. [5] demonstrated that RAG-based systems outperform traditional generative models in knowledge-intensive tasks by simplifying the search process and improving retrieval performance. [2] further enhanced the quality of generated answers by integrating large external knowledge bases, particularly in specialized fields such as medicine and law.

Shuster et al. [45] proposed a multi-task learning framework that combines RAG with multi-task learning, improving performance in tasks like knowledge-based QA and recommendation systems. The ability of RAG to reduce hallucination while increasing the relevance of generated responses has made it a key technology in modern QA systems [19]. Xu et al. [50] also explored the application of RAG in complex QA scenarios, proving its effectiveness in handling heterogeneous data sources.

2.3 Multi-Path Retrieval Approaches

While RAG has significantly reduced hallucinations, traditional single-path retrieval methods still have limitations [45]. Single-path retrieval, such as keyword retrieval, vector-based retrieval, or graph-based retrieval, only addresses one dimension of query processing and may miss critical information [21]. For example, keyword retrieval is prone to missing relevant information when the query does not exactly match the text in the document [13]. Vector-based retrieval captures semantic information but may struggle with low-frequency terms or ambiguous semantics, leading to less relevant results [4].

To tackle these limitations, multi-path retrieval methods have been proposed. Multi-path retrieval combines keyword, vector, and graph-based approaches to handle queries from multiple perspectives [5, 26, 43]. For instance, vector-based retrieval excels at semantic matching, keyword retrieval pinpoints exact expressions, and graph retrieval captures structured relationships between entities. Peng et al.[37] introduced a graph-based multi-path retrieval framework that effectively handles heterogeneous data sources, improving retrieval accuracy and efficiency.

Several notable works have demonstrated the advantages of multi-path retrieval [32]. Karpukhin et al. [22] introduced Dense Passage Retrieval (DPR), which uses dense vector representations to capture semantic information and improve retrieval performance in open-domain QA. Seo et al. [42] proposed a Real-Time QA system that combines keyword and semantic retrieval, significantly enhancing accuracy and efficiency in large-scale unstructured data.

2.4 Recent Advances in Retrieval-Augmented Generation

Recent work has focused on improving RAG by combining it with additional retrieval mechanisms. Toolformer [41] introduced a tool-enhanced retrieval framework that leverages automated tool selection to optimize retrieval results and improve the quality of generated answers. Llama2 [30, 46] and its variants, such as Ret-Llama2 and Ret-ChatGPT, have further optimized retrieval paths, leading to superior performance in complex QA tasks.

Models like CoVE [11] and SAIL [29] also leverage multi-path retrieval to significantly improve their performance in knowledge-intensive tasks. These models can dynamically retrieve relevant information from external knowledge bases, leading to more accurate and comprehensive answers. Similarly, models like Llama2-FT and Alpaca have shown advantages in handling multi-source heterogeneous data by integrating various retrieval mechanisms [46].

3 URAG

This chapter proposes a technical framework based on Unified Retrieval-Augmented Generation (RAG) and elaborates the specific computational process of the framework using mathematical symbols and formulas. By integrating three retrieval mechanisms—keyword retrieval, vector retrieval, and graph retrieval—the framework enables efficient and high-quality utilization of large-scale, multi-source, and heterogeneous data. This is especially advantageous when dealing with multi-dimensional and multi-layered information, enhancing the accuracy and comprehensiveness of the generation process.

Assuming the user's input query is q, the goal of the entire framework is to find the most relevant information from the knowledge base and generate the final answer a. As shown in Figure 1, the overall process of the framework can be summarized in the following three steps:

- **Knowledge Storage**: Constructing vector, keyword, and graph indices.
- **Knowledge Retrieval**: Extracting the most relevant documents or subgraphs based on the synergy of the three retrieval mechanisms.
- **Fusing Retrieval Results**: Merging and processing the outputs from three retrieval paths, and then the user's question and the fusing retrieval results are input into the language model to generate the final response a.

3.1 Knowledge Storage

Symbol Definition: To facilitate the description of the computational process of the subsequent modules, the following symbols are defined:

Table 1: Symbol and Definitions

Symbol	Definition
q	User query
D	Document collection in the knowledge base
d_i^c	The c-th chunk of document d_i
$v(d_i^c)$	Vector representation of document chunk d_i^c
V	The V represents the set of entity nodes
E	The E represents the set of relationship edges
$K(q)$	Set of keywords extracted from query q
$R(q)$	Set of documents retrieved based on query q
f	Similarity function used during retrieval

Vector Index: The purpose of vector indexing is to retrieve relevant documents based on semantic similarity. The process can be divided into the following steps:

- Document Chunking: Split documents D into smaller document chunks $\{d_i^1, d_i^2, \ldots, d_i^m\}$ for further processing.
- Vector Representation: Use a pre-trained embedding model $\mathcal{E}(\cdot)$ to convert document chunk d_i^c into a high-dimensional vector representation $v(d_i^c)$:
$$v(d_i^c) = \mathcal{E}(d_i^c)$$
- Storing Vectors: Store the vector representations of all document chunks in a vector database to support similarity-based retrieval later on.

Graph Index: The graph index represents entities and relationships in documents by constructing a knowledge graph.

- Triple Extraction: Extract entities and their relationships from documents to form triples (v_i, r_{ij}, v_j), where v_i and v_j represent entities, and r_{ij} represents relationships between entities.
- Graph Construction: Build the knowledge graph $G = (V, E)$ using triples, providing structured data for graph retrieval in later stages.

Keyword Index: Keyword indexing is based on inverted index technology, suitable for efficient text-matching tasks.

- Tokenization and Keyword Extraction: Use tokenization tools to segment the document D, generating a list of words $\{w_1, w_2, \ldots, w_n\}$.
- Inverted Index Construction: Build the mapping between words and document chunks, i.e., the inverted index, represented as:
$$I(w) = \{d_i^c \mid w \in d_i^c\}$$

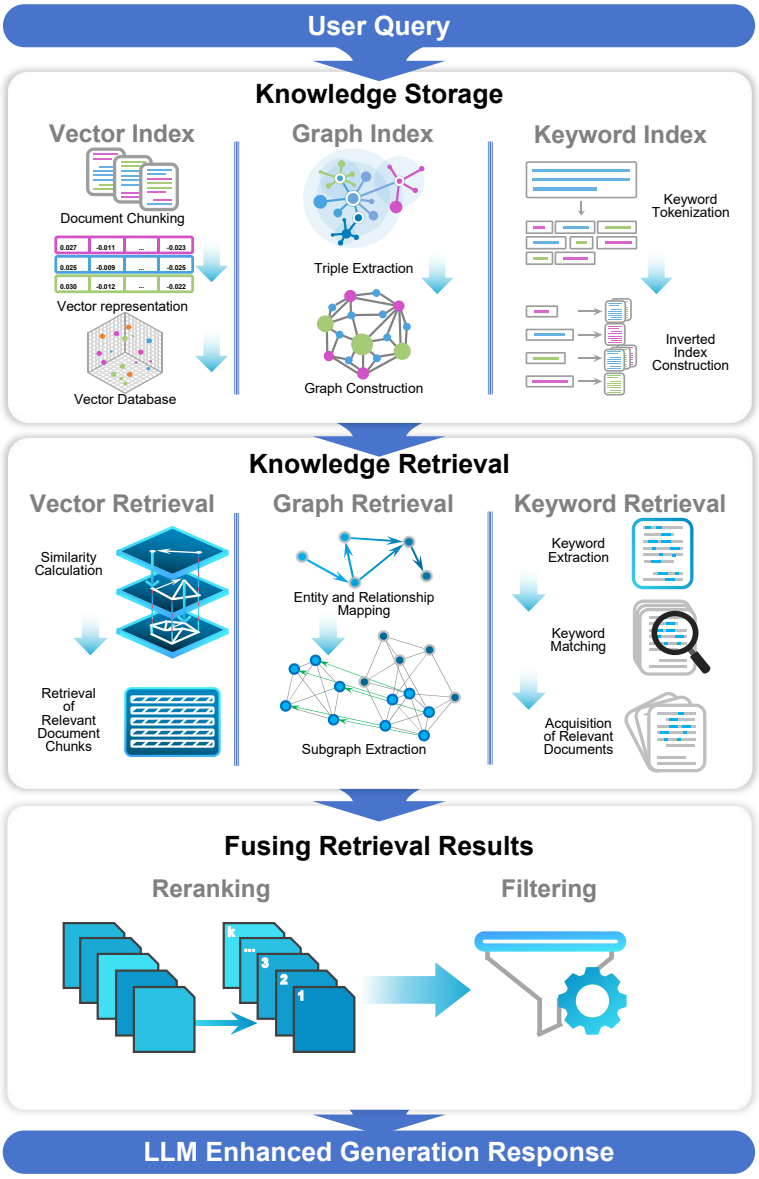

Figure 1: The architecture of URAG.

3.2 Knowledge Retrieval

The knowledge retrieval module combines vector retrieval, keyword retrieval, and graph retrieval to filter out the most relevant information from the knowledge base for user query q.

Vector Retrieval: Vector retrieval focuses on calculating the similarity between the vector representation of the query q, $v(q)$, and the document chunk $v(d_i^c)$.

- Query Vector Representation: Convert the user query q into a vector representation $v(q)$ using the embedding model $\mathcal{E}$:

$$v(q) = \mathcal{E}(q)$$

- Similarity Calculation: Calculate the similarity between the query vector $v(q)$ and the document chunk vector $v(d_i^c)$, typically using cosine similarity:

$$f(v(q), v(d_i^c)) = \frac{v(q) \cdot v(d_i^c)}{\|v(q)\| \|v(d_i^c)\|}$$

- Retrieval of Relevant Document Chunks: Rank document chunks based on similarity and return the most relevant set of chunks $R_v(q)$.

Graph Retrieval: Graph retrieval matches the query with entities and relationships in the knowledge graph, extracting the relevant subgraph.

- Entity and Relationship Mapping: First, map the keywords in the query to entities and relationships in the graph.
- Subgraph Extraction: Extract the relevant subgraph $G_q = (V_q, E_q)$ based on the entities and relationships in the query, where $V_q \subseteq V$, $E_q \subseteq E$.

Keyword Retrieval: Keyword retrieval leverages the inverted index for quick matching.

- Keyword Extraction: Extract the set of keywords $K(q)$ from the query q.
- Keyword Matching: Use the inverted index to match keywords, returning the set of document chunks containing relevant keywords:

$$R_k(q) = \bigcup_{w \in K(q)} I(w)$$

3.3 Fusing Retrieval Results

The results obtained from VectorRAG, GraphRAG, and keyword retrieval are optimized through operations such as re-ranking and filtering, and then the user's question and the fusing retrieval results are input into the language model to generate the final response.

- Reranking: Suppose the results from the three retrieval methods are $R_v(q)$, $R_k(q)$, and $R_g(q)$ respectively, and they may be ranked according to different criteria. In this paper, we use BGE Rerank (BAAI General Embedding Reranker) to rerank the initially retrieved documents, improving the relevance and accuracy of the retrieval results, yielding the final fused result $R'(q)$.
- Filtering: Remove redundant documents from the reranked result set to provide more accurate and effective contextual information for the subsequent generation step. First, pre-set deduplication thresholds or criteria based on the task's characteristics and requirements; then calculate the similarity between each pair of documents and remove those with similarities higher than the deduplication threshold, eliminating redundant information. Finally, the remaining document set after filtering becomes the final candidate set $R_f(q)$, used for the subsequent generation step.
- LLM-Enhanced Response Generation: Input the fused retrieval results $R_c(q)$ into the large-scale language model $\mathcal{M}$, generating the final response a:

$$a = \mathcal{M}(q, R_c(q))$$

4 Experimental Setup

4.1 Task Descriptions and Evaluation Metrics

To evaluate the performance of the Unified Retrieval-Augmented Generation (URAG) framework, we conducted experiments across various tasks, covering both general knowledge and specialized domains. These tasks include Short-form Question Answering (QA), Closed-set QA, and Long-form generation with citations. Each task used specific datasets and evaluation metrics, as described below:

Short-form Generation Tasks We used the PopQA [31] dataset, which contains 1,399 rare entity queries. The goal for this task was to generate accurate factual responses. The primary metric used to evaluate performance was accuracy (acc), measuring the correctness of the generated answers.

Closed-set Tasks: In tasks like Pub (PubHealth) [52] and ARC (ARC-Challenge) [7], models were required to retrieve correct answers from a fixed knowledge base. We again employed accuracy (acc) as the key evaluation metric to assess model performance.

Long-form Generation Tasks: In datasets such as Bio and ASQA[15], models needed to produce long, accurate responses supported by citations. For these tasks, we used several evaluation metrics:

- FactScore (FS): Measures the factual accuracy of the generated answers.
- Exact Match (em): Evaluates the percentage of answers that exactly match the reference answers.
- Rouge (rg): Assesses the similarity between the generated answers and reference texts, commonly used in summarization and long-form generation.
- MAUVE (mau): Evaluates the fluency and coherence of the generated text.
- Citation Precision (prec) and Citation Recall (rec): Measure the precision and recall of the citations, assessing how relevant and comprehensive the citations are in supporting the generated content.

4.2 Baselines

We compared the URAG framework against several baseline models, including non-retrieval language models and retrieval-augmented models. Some results were cited from previous work [1].

Baselines without retrievals. . We evaluate several strong, publicly available pre-trained large language models (LLMs), including Llama2 7B and 13B [46], as well as instruction-tuned models such as Alpaca 7B and 13B [12] (our replication based on Llama2). Additionally, we include models trained and fine-tuned with private data, such as ChatGPT [34] and Llama2-chat 13B. For instruction-tuned LMs, we use the official system prompt or instruction format used during training when publicly available. We also compare our method with concurrent work, CoVE 65B [11], which introduces iterative prompt engineering to enhance the factual accuracy of LLM outputs.

LMs with Retrieval: Language models with retrieval capabilities include Llama2-FT-7B [1], a fine-tuned variant of Llama2 with additional retrieval mechanisms aimed at improving task-specific performance; and SAIL-7B [29], which employs a graph-based retrieval approach to better comprehend complex entity relationships, thus enhancing both accuracy and the generation of citations.

5 Experiment Analysis

In this section, we analyze the performance of the Unified Retrieval-Augmented Generation (URAG) framework across various tasks, based on the results presented in Table 2. The evaluation covers

Table 2: Overall experiment results on six tasks. Bold numbers indicate the best performance among non-proprietary models, and gray-colored bold text indicates the best proprietary model when it outperforms all non-proprietary models. ∗ indicates concurrent or recent results reported by concurrent work. – indicates numbers that are not reported by the original papers or are not applicable. Models are sorted based on scale. FS, em, rg, mau, prec, rec denote FactScore (factuality); str-em, rouge (correctness); MAUVE (fluency); citation precision and recall, respectively.

LM	Short-form	Closed-set		Long-form generations (with citations)					
	PopQA	Pub	ARC	Bio	ASQA				
	(acc)	(acc)	(acc)	(FS)	(em)	(rg)	(mau)	(pre)	(rec)
Baselines without retrieval									
Llama2$_{7B}$	14.7	34.2	21.8	44.5	7.9	15.3	19.0	–	–
Alpaca$_{7B}$	23.6	49.8	45.0	45.8	18.8	29.4	61.7	–	–
Llama2$_{13B}$	14.7	29.4	29.4	53.4	7.2	12.4	16.0	–	–
Alpaca$_{13B}$	24.4	55.5	54.9	50.2	22.9	32.0	70.6	–	–
CoVE$_{65B}$*	–	–	–	71.2	–	–	–	–	–
Baselines with retrieval									
Llama2$_{7B}$	38.2	30.0	48.0	78.0	15.2	22.1	32.0	2.9	4.0
Alpaca$_{7B}$	46.7	40.2	48.0	76.6	30.9	33.3	57.9	5.5	7.2
Llama2-FT$_{7B}$	48.7	64.3	65.8	78.2	31.0	35.8	51.2	5.0	7.5
SAIL*$_{7B}$	–	69.2	48.4	–	–	–	–	–	–
Llama2$_{13B}$	45.7	30.2	26.0	77.5	16.3	20.5	24.7	2.3	3.6
Alpaca$_{13B}$	46.1	51.1	57.6	77.7	**34.8**	36.7	56.6	2.0	3.8
SELF-RAG$_{7B}$	54.9	72.4	67.3	**81.2**	30.0	35.7	74.3	66.9	67.8
SELF-RAG$_{13B}$	**55.8**	74.5	73.1	80.2	31.7	37.0	71.6	70.3	71.3
Our URAG$_{7B}$	53.3	74.1	78.5	69.4	31.2	37.4	42.3	70.4	73.2
Our URAG$_{13B}$	54.0	**76.2**	**80.1**	74.3	32.6	**39.0**	45.6	**72.5**	**75.4**

short-form question answering, closed-set question answering, and long-form generation tasks. The comparison includes multiple baseline models, both with and without retrieval mechanisms, highlighting URAG's advantages in terms of accuracy, retrieval quality, and generation reliability.

Short-form Question Answering (PopQA): In the PopQA task, SELF-RAG 13B achieved the highest accuracy (55.8%), marginally outperforming URAG 13B, which achieved 54.0%. However, URAG's performance remains close to the best and significantly surpasses models without retrieval mechanisms, such as Llama2 and Alpaca. For instance, Llama2 7B only reached 14.7% accuracy, showing the limitations of models that lack retrieval capabilities when handling long-tail factual queries.

Closed-set Question Answering (Pub and ARC):In the Pub dataset, URAG 13B achieved the best accuracy at 76.2%, surpassing SELF-RAG 13B at 74.5%. This indicates that URAG's multi-path retrieval mechanism is particularly effective in extracting relevant documents in specialized domains.

For the ARC dataset, URAG 13B outperformed all models with an accuracy of 80.1%, significantly higher than SELF-RAG 13B (73.1%) and other baseline models. This demonstrates URAG's superior ability to handle multi-step reasoning tasks, which are characteristic of the ARC dataset.

Long-form Generation (Bio and ASQA):In the Bio task, SELF-RAG 7B achieved the highest FactScore (FS) at 81.2%, slightly outperforming URAG 13B with 74.3%. Nevertheless, URAG still demonstrated strong performance in terms of factual correctness and generation quality.

In the ASQA task, URAG excelled in several key metrics, particularly in citation precision (prec) and recall (rec). URAG 13B achieved a citation precision of 72.5% and a recall of 75.4%, outperforming SELF-RAG 13B (70.3% and 71.3%, respectively). This indicates that URAG generates more accurate and comprehensive long-form answers with better citation support, highlighting its effectiveness in tasks that require robust citation mechanisms.

Comparison by Model Scale: Compared to smaller models such as Llama2 7B and Alpaca 7B, the larger URAG 13B and SELF-RAG 13B models significantly outperform in all tasks. This underscores the importance of combining a larger model scale with effective retrieval mechanisms to improve accuracy and generation quality.

6 Conclusion

In response to the challenges faced by question-answering technology, particularly the shortcomings in retrieval capabilities and the hallucination problem when large models generate responses, this paper proposes a knowledge question-answering framework based on Unified Retrieval-Augmented Generation (URAG). This framework integrates three retrieval mechanisms: keyword retrieval, vector retrieval, and graph retrieval, enabling the efficient and high-quality utilization of vast, multi-source, and heterogeneous data. The experimental results demonstrate that the URAG framework performs well across datasets in various task scenarios. Notably, it enhances the accuracy and comprehensiveness of knowledge retrieval, especially when dealing with multi-dimensional and multi-level information.

Specifically, the URAG framework overcomes the limitations of single retrieval paths by introducing a three-path retrieval mechanism. Keyword retrieval ensures the precise matching of keywords in user queries, vector retrieval enhances the flexibility and depth of retrieval at the semantic level, and graph retrieval further extends the capability to handle structured data and complex relationships. This multi-path collaborative approach not only improves the precision of retrieval but also enhances the comprehensiveness and traceability of the results, thereby increasing the reliability and practicality of intelligent question-answering systems.

Future work will focus on further optimizing the retrieval algorithms and generation models within the URAG framework to improve its performance in specific domains or complex scenarios. Additionally, by simply adjusting retrieval components or introducing new retrieval technologies, the URAG framework can easily adapt to different generation tasks and domain-specific requirements. As technology continues to advance, we also look forward to the emergence of more innovative retrieval-augmented generation techniques to jointly drive the development and application of question-answering technology.

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